

Graphyti — Progress Update

Status: Core procedural pipeline established (urban scene - residential, commercial, and neon zones, with a toggleable beach zone; forest scene), both built with modular components. Development is now moving into three feature areas that extend this foundation.

1. Image-to-Shader

A shader-authoring feature that replaces manual shader creation with reference-driven generation. A single input image such as a film still, photograph, or mood board is used to derive a custom shader, which is then applied uniformly across an entire scene rather than to an individual asset. Surfaces including buildings, roads, foliage, water, and sky inherit a consistent aesthetic from one reference. This reduces style iteration to an input-driven process rather than a manual authoring task.

2. Grid Expansion

Each generated scene functions as a single tile. Because scene generation is procedural and tileable, additional tiles can be generated outward from an initial seed using identical rules, producing continuous boundaries with no visible seams between tiles. World size is not constrained by this method, and tiles are streamed in on demand, keeping memory usage flat as the map expands. In addition, the underlying placement logic has been extended: where object and terrain placement was previously governed by procedural rules alone, a reference image can now inform that logic, producing organic-feeling distribution while preserving tileability.

3. Asset Generator

A complementary capability for generating individual 3D assets on demand, rather than full scenes, for example, a specific vehicle ("a weathered delivery van, right-hand drive"), a named tree species, or a piece of street furniture. Output is delivered as complete geometry, including mesh, materials, and collision data, auto-labeled and ready for direct placement in the environment.